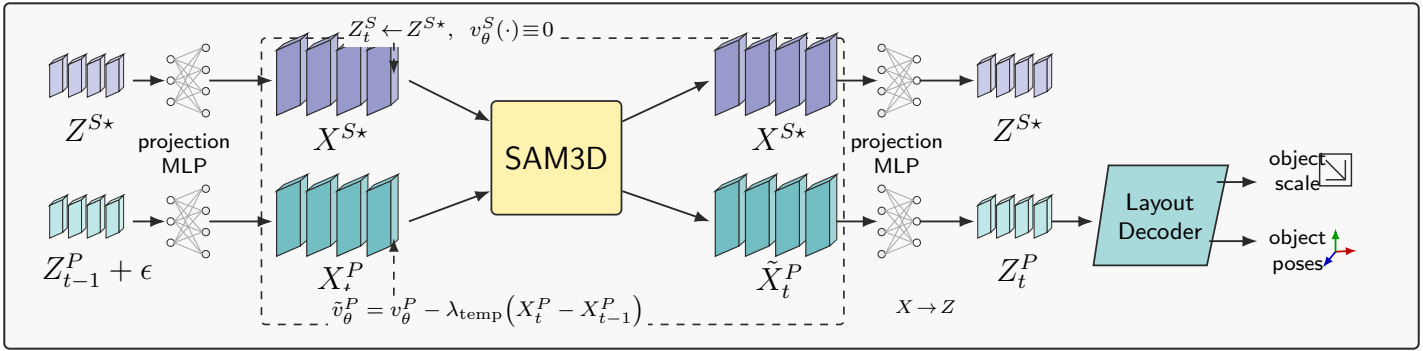

$Z_t^S \leftarrow Z^{S\star}$, $v_\theta^S(\cdot) \equiv 0$
$Z^{S\star}$
projection MLP
$X^{S\star}$
$Z_{t-1}^P + \epsilon$
X_t^P
SAM3D
$X^{S\star}$
$\tilde{X}_t^P$
projection MLP
$Z^{S\star}$
Z_t^P
$X \to Z$
Layout Decoder
object scale
object poses
$\tilde{v}_\theta^P = v_\theta^P - \lambda_{\text{temp}}\left(X_t^P - X_{t-1}^P\right)$